1️⃣ Competency 5: Interpreting and comparing multiple representations (Highest relevance)

Definition:

> Interpret mathematical outcomes in a variety of formats… compare or evaluate two or more representations interpret

🔍 How it appears in the task

- Page 2 asks students to **choose the correct graph (A–D)** representing the two payment schemes. selling-newspapers-qs3
- This involves:
  - Converting **verbal descriptions (payment rules) → graphical representations**
  - Comparing **two functional relationships** (Zedland Star vs Zedland Daily)

📌 Core insight

The essence of the task is:

- Transforming:
  - a **piecewise linear function (Star)**
  - and a **linear function (Daily)**
- Into graphical form
- Then evaluating which graph correctly represents them

👉 This is a classic case of:

> "Translation across representations + graphical interpretation + model comparison"

✔️ This is the **central interpret competency** in this problem

2️⃣ Competency 9: Understanding the relationship between context and representation

Definition:

> Understand the relationship between the context of the problem and representation interpret

🔍 How it appears in the task

- Real-world context:
  - Number of newspapers sold → income
- Mathematical structure:
  - Star: **piecewise function** (first 240 vs additional sales)
  - Daily: **linear function with intercept**

📌 Key requirement

Students must understand:

- "0.20 → 0.40" implies a **change in slope (kink in graph)**
- "60 + 0.05x" implies a **positive intercept**

👉 Without building the mapping:

> "context → function → graphical features"

students cannot identify the correct graph

3️⃣ Competency 1: Interpreting mathematical results back into context

Definition:

> Interpret a mathematical result back into the real world context interpret

🔍 How it appears in the task

Students need to interpret:

- Which line represents which newspaper
- What the intersection point means:
  - e.g., when the two jobs give equal income

📌 Deeper meaning

This goes beyond plotting:

- It requires understanding the **economic meaning of the graph**

👉 This reflects:

> "Graph → real-world interpretation"

4️⃣ Competency 2: Judging whether results make sense in context

Definition:

> Identify whether a mathematical result makes sense given the context interpret

🔍 How it appears in the task

For example:

- Star should:
  - start at 0 (no base salary)
  - become steeper after a threshold
- Daily should:
  - have a positive intercept (60 zeds)
  - have a smaller slope (0.05)

📌 Reasoning process

Students must eliminate graphs with:

- incorrect slopes
- incorrect intercepts
- missing "kink" (for piecewise function)

👉 This is essentially:

> "sense-checking graphical representations"

5️⃣ Competency 7: Constructing explanations and arguments

Definition:

> Construct and communicate explanations interpret

🔍 How it appears in the task

Although it is multiple choice, it implicitly requires:

- explaining why one graph is correct
- explaining why others are incorrect

📌 Educational perspective

Students should be able to justify:

- slope differences
- existence of a breakpoint
- intercept meaning

👉 This reflects an **implicit reasoning and explanation ability**

1️⃣ 能力5：多种表征之间的解释与比较（最高相关）

对应条目：

> Interpret mathematical outcomes in a variety of formats… compare or evaluate two or more representations interpret

🔍 题目体现

- 第2页要求：**从A–D四个图中选出正确图像表示两种报酬模式**
- 涉及：
  - **文字描述（支付规则） → 图像（函数关系）转换**
  - **两种函数（Zedland Star vs Daily）之间的比较**

📌 本质分析

该题核心是：

- 将**分段线性函数（Star）与线性函数（Daily）**
- 转换为**图像表达**
- 并判断哪一个图正确反映两者关系

👉 属于典型的：

> “不同表征之间转换 + 图像判断 + 多模型比较”

✔️ 这是本题最核心的 interpret 能力

2️⃣ 能力9：情境与数学表示之间的关系理解

对应条目：

> Understand the relationship between the context of the problem and representation interpret

🔍 题目体现

- 现实情境：
  - 卖报纸数量 → 收入
- 数学结构：
  - Star：**分段函数（前240份 vs 之后）**
  - Daily：**固定底薪 + 线性增长**

📌 关键点

学生必须理解：

- “0.20 → 0.40”意味着**斜率变化（折线）**
- “60 + 0.05x”意味着**截距不为0**

👉 若不能正确建立：

> “现实规则 → 函数结构 → 图像特征”

则无法选对图

3️⃣ 能力1：将数学结果解释回现实情境

对应条目：

> Interpret a mathematical result back into the real world context interpret

🔍 题目体现

学生需理解：

- 哪条线代表哪个报社
- 交点意味着：
  - “在哪个销量下收入相同”

📌 深层体现

不仅是画图，还要理解：

- 哪种方案更赚钱（随销量变化）
- 图像的经济含义

👉 属于“图像 → 现实意义”的反向解释

4️⃣ 能力2：判断结果是否符合情境

对应条目：

> Identify whether a mathematical result makes sense given the context interpret

🔍 题目体现

例如：

- Star 应该：
  - 起点在0（没有底薪）
  - 中途“变陡”（单价提高）
- Daily 应该：
  - 有正截距（60 zeds）
  - 斜率较小（0.05）

📌 判断逻辑

学生需要排除：

- 错误斜率
- 错误截距
- 没有“折点”的图

👉 本质是：

> “合理性检验（graph sense check）”

5️⃣ 能力7：解释与论证

对应条目：

> Construct and communicate explanations interpret

🔍 题目体现

虽然题目是选择题，但隐含要求：

- 为什么某图正确？
- 为什么其他图错误？

📌 教学层面

学生应能说明：

- 哪条线斜率更大
- 为什么有折线
- 为什么有截距

👉 属于“隐性解释能力”